

Discovery Executive Summary

Project: discovery12345 · Generated: 7/17/2026, 1:27:52 PM

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating	Hotspot Score
1	Architecture & Design Analysis	HIGH RISK	—

1. Architecture & Design Analysis

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by High-Risk cross-domain coupling (H8), legacy React Router v5 patterns (F5), and duplicated domain logic in view components (H10).

EXECUTIVE SUMMARY

This TARGET_WORKSPACE is a **frontend-only** social-network SPA (`social-media-react`) with **90 source files** (12 pages, 45 components, 11 services, 4 Redux modules). **No backend/server layer exists in the repository** — all persistence goes through an external REST API at `localhost:3030/api/` (production: `/api/`). Architectural health is **mixed**: HTTP access is partially centralized via `httpService` and domain service modules, but **presentation components bypass Redux** in 16 files, **business rules are duplicated** across post/comment/reply components, and **four feature domains** (user, post/comment, chat, activity) are tightly coupled through shared Redux state, socket orchestration in `Main.jsx`, and direct cross-service calls. The dominant risks are **legacy React Router v5 patterns (20 files)**, **cross-domain coupling (8 access points)**, and **inconsistent data-flow conventions** that will amplify change cost as features grow.

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	N/A (no backend)	GOOD
H2	Missing Service Layer	Controllers accessing repos/models	<10	10–20	>20	0 backend handlers	GOOD
H3	Missing Repository Pattern	Direct DB access points	<10	10–20	>20	0 (external API)	GOOD
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0 cycles (88 files, 137 edges)	GOOD
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	0 (<code>utilService</code> is generic)	GOOD
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	N/A (no backend)	GOOD
H7	God Classes	Classes >1000 LOC	0	1–3	>3	0 (max: <code>postActions.js</code> 210 LOC)	GOOD
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	8	HIGH RISK
H9	Shared Database Coupling	Tables shared across domains	<10%	10–30%	>30%	N/A (schema external)	GOOD
F1	Business Logic in Components	Avg LOC per component	<150	150–300	>300	78.6 avg (58 <code>.jsx</code> files)	GOOD
F2	Missing Frontend Service/Data Layer	Components w/ inline API calls	<10	10–20	>20	16	MODERATE
F3	God / Oversized Components	Components >400 LOC	0	1–3	>3	0 (max: <code>Message.jsx</code> 250 LOC)	GOOD
F4	Prop Drilling / Global State Abuse	Max prop-drilling depth	≤2	3–4	>4	2 levels; 71 <code>useSelector</code> hooks	MODERATE
F5	Legacy / Inconsistent Component Patterns	Legacy-pattern components	0	1–10	>10	20	HIGH RISK
H10	Duplicated Domain Logic	Components duplicating	0	1–3	>3	5	MODERATE

	in Views (additional)	reaction/connection rules				
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1.4 Actions Required

Hotspot	Action	Rating	Priority
H8	Split cross-domain imports: move comment actions out of <code>postActions.js</code> , add <code>activityFacade</code> for notification DTO assembly, decentralize socket subscriptions from <code>Main.jsx</code>	HIGH RISK	CRITICAL
F5	Migrate React Router v5 (<code>HashRouter</code> , <code>component=</code> , <code>useHistory</code>) to v6 across 20 files; standardize service naming	HIGH RISK	HIGH
F2	Eliminate 16 direct service imports from components — route all reads through Redux thunks and selectors	MODERATE	HIGH
F4	Introduce <code>AuthContext</code> + memoized selectors; document single data-access convention	MODERATE	MEDIUM
H10	Extract shared <code>toggleReaction</code> and <code>addConnection</code> domain helpers; remove duplicated logic from 5 view files	MODERATE	MEDIUM

1.5 Expected Outcomes

- **Bounded feature modules** — user, post, comment, chat, and activity can evolve independently with explicit facades instead of cross-imports.
- **Single data-flow path** — components consume selectors/thunks only, eliminating dual Redux + direct-service paths and stale cache bugs in `userService.usersCash`.
- **Testable domain rules** — extracted reaction/connection helpers become unit-testable without mounting React components.
- **Lower migration cost** — React Router v6 upgrade unlocks modern layouts, error boundaries, and code-splitting patterns.
- **Immutable, predictable state** — normalized Redux store removes embedded comment graphs and direct prop mutation anti-patterns.